Yutian Qi
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EDUCATION

Nanjing University **Nanjing, China**
B.Sc. in Atmospheric Science (Dual Degree with University of Helsinki) **Sep 2023 – Jul 2027**
Core courses: *Scientific Computing – Programing Language & Numerical Methods, Data Structure and Algorithms, Statistics and Data Analysis in Earth Science, Introfuction to Artificial Intelligence, Introduction to Atmospheric Science, Introduction to Evironmental Science, etc.*

TECHNICAL SKILLS

Programming Languages: Python, C/C++, JavaScript, Swift; SQL
Data Analysis & Machine Learning: NumPy, Pandas; Matplotlib, xarray, PyTorch, TensorFlow, SciPy, Scikit-learn
Agent & Multimodal AI: RAG, MCP Server
Web Development: Node.js, Express, Flask, FastAPI; PostgreSQL, Vue3, Vite
Developer Tools: Git, Docker; Linux, Jupyter Notebook, VS Code
Office Tools: Adobe Photoshop, Microsoft Office (Word, Excel, PowerPoint);

ACADEMIC PROJECT

Multi-Object Grasping, Sparse Reinforcement Learning and Skill Transfer in Robotic Manipulation **Jun 2026 – present**
Embodied Intelligence and Robotic Systems (EMBODICA) Lab | Advisor: Zhenshan Bing

- Used LIBERO, robosuite, and MuJoCo to conduct a single-object grasping reinforcement learning in a six-object cluttered environment, including task environment configuration, reward function development, policy training, and performance evaluation.
- Developed an event-driven sparse reward state machine, dividing grasping tasks into 7 stages and designing reward functions based on gripper states, object movement, and task states to reduce hovering, pushing, and incorrect release behaviors caused by distance-based rewards.
- Used MuJoCo state snapshots for reverse curriculum training, applying staged resets, progressive difficulty adjustment, and targeted sampling to improve exploration in long-horizon tasks; increased deterministic success rates in difficult stages to 90%–100% and improved effective replay coverage for key release actions from 4.21% to 89.2%.
- Conducted multi-seed experiments to analyse the effects of actor/critic update frequency, optimizer momentum, and discount factor on policy performance, reducing incorrect release actions from 3/3 seeds to 0/3 seeds; further analysed skill transfer failures using 400 frozen-policy counterfactual experiments.
- Developed an SAC policy transfer and continual training mechanism supporting full_resume, network_warm_start, and actor_only_eval initialization modes; restored and validated actor, twin critics, target critic, optimizer states, curriculum states, and random number generator states for controlled parameter transfer and experimental comparison across object manipulation tasks.

ClimateNet-Bench: A Spatio-Temporal Machine Learning Benchmark for Land Surface Evaporation Anomaly Forecasting
https://github.com/Drink8Water/ClimateNet-Bench **Apr 2026 – Aug 2026**

- Developed a monthly land surface evaporation anomaly forecasting dataset using corrected ERA5-Land data, containing 5.318 million grid records from the Sahara and East China regions during 2019- 2023, collected 4.786 million samples with six-month lagged variables.
- Conducted preprocessing with leakage prevention, fitting monthly climatology, anomaly features, event thresholds, and normalization parameters on training data to avoid data leakage during temporal forecasting and cross-region evaluation.
- Developed evaluation protocols including random split, temporal holdout, multi-seed, and repeated region-stratified spatial folds, and conducted model evaluations under random split, future-year prediction, and cross-region transfer scenarios.
- Conducted comparative evaluation of predictive models including Linear Regression and LightGBM, comparing prediction performance using MAE, RMSE, R^2, and Skill Score; LightGBM achieved RMSE values of 5.766 ± 0.020, 6.309 ± 0.015, and 7.066 ± 1.093 under random split, temporal holdout, and repeated spatial evaluation.
- Conducted error analysis and regional performance comparison, finding that LightGBM outperformed Linear Regression across various spatial folds with a 27.8% average RMSE reduction; further analysis showed higher prediction errors in East China than Sahara across all 18 corrected regional comparisons.
- Developed a FastAPI + PostgreSQL backend and Vue 3 + ECharts visualisation platform for benchmark metadata management, leaderboard, prediction artifacts, and result visualisation

AirfoilCFD-ML: A Scientific Machine Learning Framework for 2D Airfoil Flow Prediction
https://github.com/Drink8Water/airfoil-cfd-ml **Apr 2026-Jul 2026**

- Developed neural surrogate models for 2D airfoil flow prediction using PyTorch, including SimpleCNN, ResUNet, FNO2D, GeoFNO-lite, and Transolver-lite.
- Compared multiple geometry representations, including binary masks, coordinate encoding, Signed Distance Fields (SDF), and Boundary Distance, through controlled ablation studies to analyse their impact on model prediction performance.

- Analysed model performance using Relative MAE, Divergence Error, Vorticity Error, Boundary Error, Wake Error, and Spectral Error to evaluate prediction quality from numerical, physical, and frequency-domain perspectives.
- Managed model configurations, checkpoints, training logs, and evaluation results, enabling benchmark comparisons across different model architectures and geometry representations.
- Conducted 200+ automated tests with pytest for data processing, models, loss functions, and evaluation modules, improving code maintainability and experiment reproducibility.

Apple Vision Pro Immersive Astronomy Data Visualization System **Nov 2024 – Dec 2025**

Project Leader | Undergraduate Innovation and Entrepreneurship Training Program

- Used Astropy and NumPy to process FITS images, WCS coordinates, and star catalog data, including data cleaning, coordinate transformation, missing value handling, and field standardization.
- Used Pandas for analysis of star catalog data, including object position, distance, and brightness distributions.
- Converted astronomical observations (RA, Dec, and Distance) into 3D point clouds, and adjusted colour, transparency, and particle size for visualization.
- Applied spatial partitioning, importance sampling, density control, and LOD techniques to optimise large-scale point cloud and particle data for real-time rendering.
- Used USDZ and JSON metadata to manage 3D models and point cloud data for efficient dataset loading and visualisation.
- Developed a visionOS spatial prototype using SwiftUI, RealityKit, and ARKit for 3D model browsing, layer switching, celestial object queries, etc.

PROFESSIONAL EXPERIENCE

Shanghai Fengling Technology Co., Ltd. **Jun 2026 – Present**

Research Intern (Vision-Language-Action & Embodied AI) **Remote**

- Applied Vision-Language-Action models including OpenVLA, RT-2 and π0 for visual inputs, language instructions and robot action sequences.
- Processed Observation, Action and Trajectory data from robotic manipulation tasks, including data format inspection, task information organisation and action representation analysis.
- Used PyTorch and Transformers to examine model inputs and outputs, inference and action prediction results in OpenVLA systems.
- Evaluated robotic manipulation tasks using metrics such as Success Rate and Rollout Evaluation, and reviewed failure cases.
- Participated in OpenVLA model reproduction, including training configuration preparation, inference execution and result recording.
- Compared action prediction results across different tasks and reviewed model performance on long-horizon and cross-task manipulation tasks.

China Mobile Group Jiangsu Co., Ltd., Zhenjiang Branch | AI Center of Excellence (AICoE) **Jul 2026– Aug 2026**

AI Full-Stack Development Intern **Zhenjiang, China**

- Participated in the development of an internal file management and AI assistant system, working on frontend features, backend APIs, and RAG modules; the system is currently used by 100+ users for file upload, retrieval, sharing, summarization, and document question answering.
- Developed frontend functionalities using Vue 3, TypeScript, Vite, Pinia, and Vue Router, implementing personal file spaces, public file lists, sharing inboxes, permission configuration, administration pages, and AI assistant portal.
- Developed API Client, route guards, and authentication state management logic, integrating with Go/Gin backend authentication and file management APIs; handled HttpOnly Cookie-based sessions, API exceptions, permission control, and page state management.
- Participated in AI-RAG and asynchronous document enrichment module development, implementing post-upload document processing for TXT, Markdown, DOCX/DOC, CSV, and XLSX files, including text extraction, chunking, and source location tracking; used local Qwen3.6-35B to generate summaries and keywords for retrieval enhancement..
- Developed RAGService and permission-aware Agent APIs, supporting document search, file information retrieval, original chunk access, and index rebuilding; packaged related capabilities into MCP tools including search_documents, summarize_file, ask_documents, and reindex_file, enabling integration with Web AI, MCP Server, and local Agent Skills.
- Participated in PostgreSQL schema design and migration development, maintaining RAG-related tables including ai_documents, ai_document_chunks, and ai_document_summaries, with metadata tracking for checksum, chunk index, page number, character range, processing status, and error information.

HONOURS & AWARDS

- Meritorious Winner (H Award), MCM/ICM, 2025
- Second Prize, Nanjing University Challenge Cup Competition, 2025
- People's Scholarship, Nanjing University, 2024–2025
- People's Scholarship, Nanjing University, 2023–2024